

Model Provider	Model	Release Date	Cutoff Date	Cost of Tokens	Context Window	Type
OpenAI	GPT-5.1	Nov 2025	–	\$3 / \$15 per 1M	1M	Closed
	GPT-5.2	Dec 2025	–	\$4 / \$20 per 1M	1M	Closed
	GPT-5.3 Codex / Spark	Feb 2026	–	\$2 / \$10 per 1M	512K	Closed
	GPT-5.4 (frontier + mini/nano)	Mar 2026	–	\$5 / \$25 per 1M	1M	Closed
	GPT-5.5 / Pro	Apr 2026	–	\$5 / \$25 per 1M	1M	Closed
	GPT-5.6 Sol/Terra/Luna	Jun 2026	Jul 2026	\$5 / \$30 per 1M	1.1M	Closed
	GPT-Rosalind	Jun 2026	–	\$4 / \$20 per 1M	512K	Closed
	GPT-Live	Jul 2026	–	Subscription	Voice-first	Closed
Anthropic	Claude 3.5 Sonnet	Jun 2025	–	\$3 input / \$15 output per 1M tokens	200K	Closed
Anthropic	Claude 3.5 Haiku	Jul 2025	–	\$1 input / \$5 output per 1M tokens	200K	Closed
Anthropic	Claude 3.5 Opus	Aug 2025	–	\$5 input / \$25 output per 1M tokens	200K	Closed
Anthropic	Claude Opus 4.8	May 2026	–	\$5 input / \$25 output per 1M tokens	1M	Closed
Google DeepMind	Gemini 2.0 Ultra	Dec 2025	–	\$2 input / \$12 output per 1M tokens	1M	Closed

<u>Google DeepMind</u>	Gemini 2.0 Pro	Dec 2025	–	\$1.50 input / \$8 output per 1M tokens	512K	Clo
<u>Google DeepMind</u>	Gemini 2.0 Nano	Dec 2025	–	\$0.50 per 1M tokens	128K	Clo
<u>Google DeepMind</u>	Gemini 3.0	Jan 2026	–	\$2.20 input / \$13 output per 1M tokens	1M	Clo
<u>Google DeepMind</u>	Gemini 3.1 Pro	Feb 2026	Feb 2026	\$2.50 input / \$15 output per 1M tokens	2M	Clo
<u>Microsoft</u>	Phi-3.5	Oct 2025	–	\$1 input / \$5 output per 1M tokens	256K	Clo
<u>Microsoft</u>	Phi-4	Apr 2026	–	\$1.20 input / \$6 output per 1M tokens	512K	Clo
<u>DeepSeek</u>	DeepSeek V3	Sep 2025	–	\$1 input / \$2 output per 1M tokens	512K	Op we
<u>DeepSeek</u>	DeepSeek V4-Pro	Apr 2026	–	\$1.74 input / \$3.48 output per 1M	1M	Op we

				tokens		
Mistral AI	Mixtral 8x22B	Dec 2025	–	\$1.50 per 1M tokens	256K	Op we
Mistral AI	Mistral Large 2	Mar 2026	–	\$2 input / \$10 output per 1M tokens	256K	Op we
Perplexity AI	Perplexity Pro Search	Jun 2026	–	Subscription-based (no per-token pricing)	128K	Clo
Perplexity AI	Perplexity Answer Engine (upgrade)	Jul 2026	–	Subscription-based	128K	Clo
xAI	Grok 3.5	Nov 2025	–	\$2 per 1M tokens	256K	Clo
xAI	Grok 4.3	Mar 2026	–	\$2 per 1M tokens	512K	Clo

Meta	Llama 3.1	Jul 2025	–	Free (open-weight)	128K	Open
Meta	Llama 4 Scout	Apr 2026	–	~\$0.80 per 1M tokens	~500K (claimed 10M)	Open
Sarvam AI	Sarvam Indic-LLM	May 2026	–	\$0.25 per 1M tokens	128K	Open

Work-Mechanism of Transformers:

1. Self-attention mechanism:

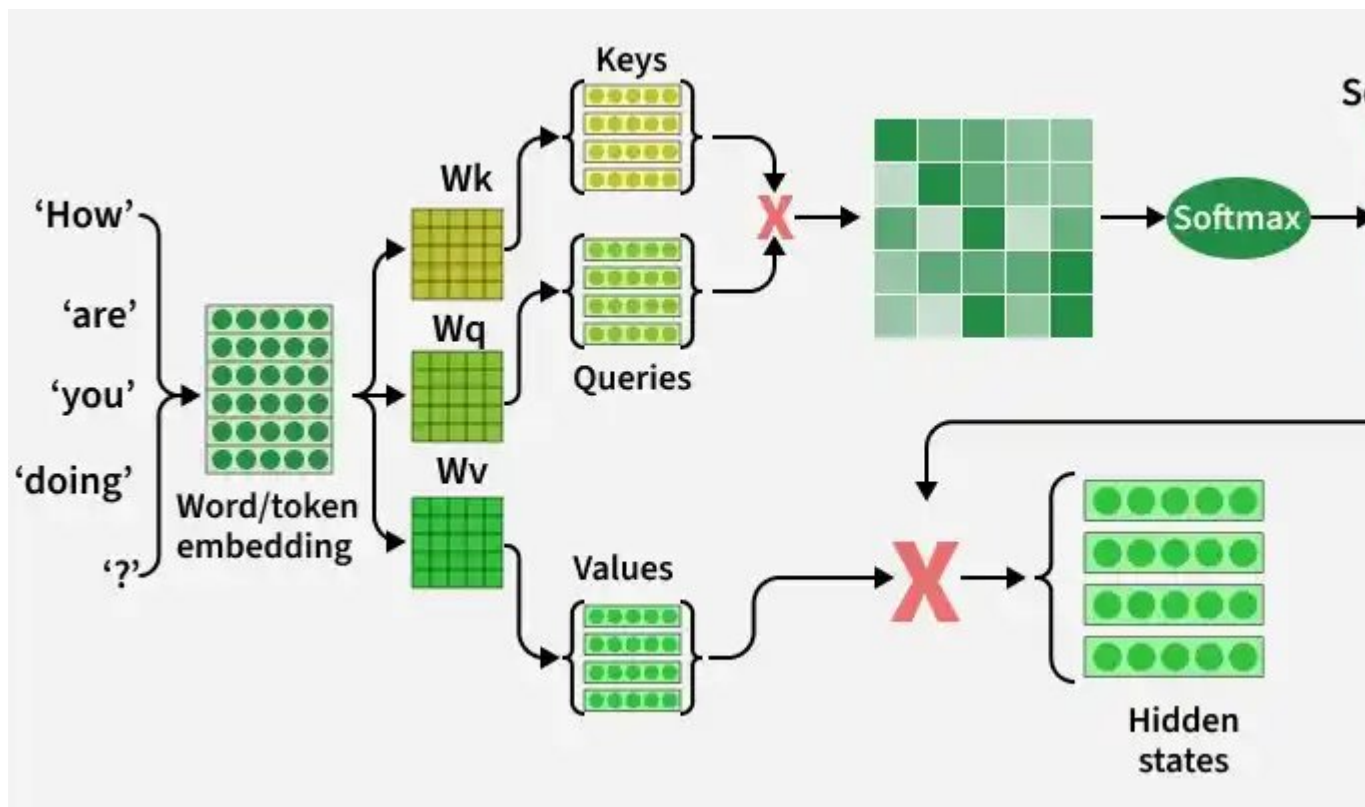
Self-attention provides a dynamic approach to generating word representations by considering the entire context of the input sequence. Instead of relying on static embeddings, self-attention adjusts the representation of each word based on the surrounding words, effectively creating contextual embeddings.

self-attention mechanism operates on three fundamental components: Queries (Q), Keys (K), and Values (V).

Queries (Q)----- The Query is essentially the representation of the word (or token) that the model is currently focusing on.

Keys(K) -----The Keys represent all the words in the sequence, including the word currently being focused on. Each word in the sequence has a corresponding key vector.

Values(V)----- The Values are what the model ultimately uses to construct its understanding of the sequence. Each word in the sequence is associated with a value vector, which holds the contextual information or the “meaning” of the word. Once the model has determined how much attention to give to each word (based on the comparison between the queries and keys), it uses the value vectors to build a weighted representation of the context.



2. Vector Embedding:

Vector embeddings are numerical representations of data points that express different types of data, including nonmathematical data such as words or images, as an array of numbers that machine learning (ML) models can process.

An embedding is any numerical representation of data that captures its relevant qualities in a way that ML algorithms can process. The data is embedded in n-dimensional space.

3. Feed forward Neural Network

The way that data moves through the architecture of the network defines Feedforward neural networks. Every neural network starts with an input and results in an output. Between those first and final steps, a neural network has hidden layers that the user can't see. The hidden layers provide the structure for how the neural network will interact with the data. At the same time, the nature of the system allows artificial intelligence to draw conclusions independent of human intervention.

In a Feedforward neural network, each node connects to the next layer's node. The data flows forward constantly, from one layer to another, with no loops or cycles. The data only feeds forward, which is where the neural network got its name.

This technology allows artificial intelligence to process data found in images or video. Eg: self driving cars, Content moderation, Manufacturing, Image search